**Assignment 1**

**Explore following 4 websites and answer the questions**

- **bitcoin.org**
- Blockchain.com
- https://coinmarketcap.com/
- https://www.investopedia.com/

# One-Word Answerable Questions from bitcoin.org

### Bitcoin Basics

1. Bitcoin was introduced by whom?
   **Satoshi Nakamoto**

2. Bitcoin transactions are recorded on which data structure?
   **Blockchain**

3. Bitcoin uses which consensus mechanism?
   **Proof of Work**

4. The smallest unit of Bitcoin is called?
   **Satoshi (sat)**
   **1 Bitcoin = 100,000,000 Satoshis**

5. Bitcoin is a **peer-to-peer** type of what system?
   **Electronic cash**

### Wallets

6. A wallet stores the user's Private key.

7. The recovery phrase used in wallets is called a Seed phrase.

8. Hardware wallets store keys <u>offline </u>(online/offline).

9. Bitcoin addresses are derived from <u>Public </u>keys.

10. What is required to sign a transaction? (public/private)
**Private**

## Security

11. Bitcoin prevents double-spending using <u>blockchain</u>.

12. Bitcoin is secured using <u>asymmetric </u>cryptography (asymmetric/symmetric).

13. To run a Bitcoin node, the software required is Bitcoin <u>core</u>.

14. The process of verifying transactions is called <u>validation</u>.

15. The Bitcoin network is described as <u>decentralized </u>(centralized/decentralized).

## Mining

16. Miners solve <u>cryptographic </u>puzzles to add blocks.

17. Mining rewards a block with newly created <u>bitcoins</u>.

18. The block time target for Bitcoin is <u>10</u> minutes.

19. Mining difficulty adjusts every **2016** block.

20. Miners compete to find a valid block <u>hash</u> (hash/signature).

## Blockchain Structure

21. Each block contains a <u>Merkle </u>root.

22. Blocks are linked using the previous block's <u>hash</u>.

23. A Bitcoin transaction contains inputs and <u>outputs </u>.

24. Unspent transaction outputs are called <u>**UTXOs** (Unspent Transaction Outputs)</u>.

25. The first-ever Bitcoin block is called the <u>Genesis </u>block.

## Network

26. A full node stores the entire blockchain Ledger.

27. Bitcoin uses which protocol for communication? (TCP/P2P)
    **TCP**

28. A light wallet uses SPV mode (SPV/full).

29. Nodes relay information across a Peer-to-peer network.

30. The maximum Bitcoin supply is 21 million.

**Task B1: Explore a Block**

Note the following from the **latest block**:

- Block height: **960833**
- Block hash:
  **00000000000000000000031f27be35daba44eecd10b82776d82f77b8e80e976c7**
- Timestamp: **2026-08-03 12:01:30 GMT +5.5**
- Number of transactions: **5516**
- Miner/pool name: **unknown**

## Task B2: Explore a Transaction

Open any one Bitcoin transaction.

10. Write the TXID of the transaction.
    **f4c76d17dc68e482e84bb18248d705fc3d4e6ab55dfeead68a309610edf26c17**

11. Number of inputs: **1**

12. Number of outputs: **1**

13. Transaction fee: **0.00000110 BTC (1.01 sat/vB)**

**14.** How many confirmations does it have?
    **960,835**

### Task B3: Explore an Address

15. Search for any Bitcoin address shown in the block.
    **bc1qylrcfj22ff8hv8szj4d3j0aqyzrwm9gl0x8z35**

16. Record:

- Total BTC received: **51.86941224 BTC**

- Total BTC sent: **51.86941224 BTC**

- Current balance: **$0.00**
  What is the role of a full node?

List any two developer resources available on bitcoin.org.
Developer Guides: Covers topics such as Blockchain, Transactions, Wallets, Mining, P2P Network, Payment Processing, and Contracts.

Reference Documentation: Provides technical details including the RPC API Reference, Blockchain Reference, Transaction Reference, Wallet Reference, and P2P Network Reference.

## Explore : https://coinmarketcap.com/

### Task C1: Market Information

- Current price of BTC: **$62,716.48**

- 24-hour % change: **1.12%**

- Market capitalization: **$1.25T**

- Circulating supply: **20.06M BTC**

- Total/maximum supply: **20.06M BTC**

- ### Task C2: Historical Data
- What was the price of Bitcoin one year ago? **$113,730**

- Note the highest and lowest prices in the last 30 days.
  **Highest Price: ~$66,700 USD**
  **Date: Late July 2026** (specifically around July 23–25, 2026)

**Lowest Price: ~$58,000 USD**
**Date: Early July 2026** (recorded around July 3–5, 2026 during the early July)

## Explore https://www.investopedia.com/

Why is Bitcoin considered volatile?
Bitcoin is considered **volatile** because its price can rise or fall significantly in a short period due to factors such as market demand and supply, investor sentiment, news, regulations, and macroeconomic events.

List two investment risks mentioned.
**High price volatility** leading to potential financial losses.
**Regulatory uncertainty**, where changes in government policies or laws can affect Bitcoin's value and usage.

- Part E: Reflection & Analysis

Which website gave you the most clear understanding? Why?
**Bitcoin.org** gave the clearest understanding because it provides well-structured explanations, beginner-friendly guides, and official documentation on Bitcoin, wallets, mining, transactions, and blockchain concepts with simple language and practical examples.

In one paragraph, explain how blockchain ensures trust without a central authority.
Blockchain ensures trust without a central authority by storing transactions in blocks that are cryptographically linked using hashes. Every transaction is verified by a decentralized network of nodes through a consensus mechanism such as Proof of Work. Once a block is added to the blockchain, altering its data would change its hash and invalidate all subsequent blocks, making tampering extremely difficult. Since every full node maintains a copy of the blockchain and independently verifies transactions, no single organization or individual controls the network, ensuring transparency, security, and trust among participants.